

Checkpoint F2: div and curl

Consider the vector field $\vec{F}(x, y, z) = \langle 7x^2, -9(x+y)^2, (x+y+z)^2 \rangle$.

Two of these things make sense to do and two of them do not. Compute the two that do make sense, and explain why the other two don't make sense.

1. $\text{div}(\text{div}(\vec{F}))$
2. $\text{div}(\text{curl}(\vec{F}))$
3. $\text{curl}(\text{div}(\vec{F}))$
4. $\text{curl}(\text{curl}(\vec{F}))$